ActInf Textbook Group ~ Cohort 6 ~ Session 2 (Chapter 1, Part 1) ~ 2/5/2024

Source: YouTube **Playlist**: Active Inference Textbook Group ~ Cohort 6 **Duration**: 56:12 | **Uploaded**: Unknown **Transcript method**: auto_caption

---

all right thank you everyone today is February 5th 2024 and Oli will be facilitating this discussion on chapter one for cohort 6 so thanks Oli okay uh hello everyone thanks for joining for our first discussion for chapter one uh well um as you might have read in in the content and uh also for the introduction in chapter one uh this is sorry this chapter gives a kind of um overall picture of uh what kind of theory active inference is uh and specifically uh what kind of approach uh the textbook is going to take explaining different aspects of um active inference Theory uh well one of the main themes of uh this chapter and U also the overall organizational scheme of the textbook is uh the difference between two kind of approach to active inference literature namely uh the high road and the low road so basically uh for high road uh the prior assumption is the existence of quote unquote things and then explaining from that um seemingly obvious assumption into what uh what requirements that quote unquote thing uh should have in order for it to have U many various uh aspects such as agency uh as a kind of decision maker and so on so basically uh it's not something specific to cognitive agents but uh the whole framework can also be applied to many other uh uh many other agents uh as long as it can be described in terms of its internal and external Dynamics so uh that's one approach to active inference the other one uh which uh is usually called Low Road in the literature is beginning from uh some of the concepts developed uh from back in the 19th century uh for example by people such as Helm holes uh in which they kind of they it tries to generalize uh the basian reasoning uh into um unto un developing uh the notion of free energy and specifically variation free energy uh for which we can um we can formulate uh the way that that the organisms or agents can perceive the world and also uh the related variable or related so-called parameter uh so for uh for perception we need a kind of lower boundary which is the AAL free energy and uh for um for control or for decision making uh we need uh another uh variable or parameter namely expected free energy so by describing these two parameters as kind of lower bound of certainty we can uh again formulate how uh organisms or cognitive agents can engage in both perception and also action so basically when we say active inference it's uh it implies a kind of integration of these two um these two aspects of perception and action so they're not uh essentially distinct from each other uh both of them uh both of them kind of um follow a similar trajectory or similar mathem iCal uh technology but uh in somewhat mirror or uh reversed way so um I hope I uh I could give a kind of overall meaning of what high road and low road uh mean but obviously we'll uh we'll get to see much more detail of both of these approaches so the uh the chapter two deals with um the introdu uction of low road approach and then beginning from chapter 3 we'll see how high road approach differs from uh the low road one so um if anyone has any thought or comments or question on these two concepts or we can go to the question section and see uh and discuss some of the questions positive there okay so maybe we can open the question section or by the way yeah no on that question um so in the in the chapter it talks about the fact that organisms or just agents can not the motivation is to reduce surprise but they may not able to act on surprise directly but instead on what's called proxies could you and they see that's a variation the free end variation free energy could you clarify what we mean by that or just giv some examples of that sure so yeah the concept of surprise uh actually is a little bit different than uh the psychological conception of surprise uh that we're familiar with in everyday parlament but uh basically what we mean inactive inference by surprise is its uh statistic in its a statistical sense so uh it's surprisal as a functional uh and uh yes so variational free energy is a kind of upper bound of this function or this surprisal this parameter uh in which uh so the the agent um when it engages in the act perception it tries to minimize this upper bound or at least uh to somehow optimize its error uh so that the uh KL Divergence of its error wouldn't exceed this uh bound to this variational free energy so basically uh by specifying this uh oper bound or or this bound as an optimization parameter uh we can deal with approximate basian inference instead of exact basian inference so we don't need to exactly or precisely specify uh the Divergence of uh the perception from uh the input data uh instead we can specify a kind of uh bound to to approximately uh minimize that error in order uh for the perception to have some meaningful um to have any kind of meaning uh for us uh in terms of its action or any uh subsequent um subsequent message passing and so on uh so yes here uh variation free energy again is a paramet is an optim optimization parameter uh so uh the agent in active inference framework uh will try to somehow minimize this uh or actually optimize uh this parameter based on the Divergence more specifically the KL callback lier Divergence between the input data and its uh generative model we'll see how it works uh more precisely and More rigorously in chapter two but uh free energy yes is the surprisal or the parameter uh the agent tries to minimize and variational free energy is kind of a get around uh so that instead of engaging in uh exact basian inference we just need to do the approximate basian inference which is much more tractable than the exact basian inference because as we know basian inference uh for U anything more than very very simple cases are extremely intractable I had a a quick question about that if that's okay yes sure um and by the way it's okay if we need to put a pin in this and um and perhaps address it in a later chapter as well I am I am interested in um not being super familiar with um with the that intractability problem do you know what it is that makes that straight up basian computation in intractable and and how does uh variational free energy get around that or should we save that for later time okay so a preliminary answer would be uh so in order to do um an exact basian inference we need to account for every single parameter every single variable that uh that is relevant uh for U any kind of real real world situation uh so as we know uh when when we want when we want to uh want to model any any kind of real world situation uh basically the sheer number of variables we need to account uh are is extremely uh I mean uh say for some very very simple cases in other cases we need to account for many many more variables so it makes uh the situation much more complicated uh so um and also we might not also know uh the exact Val for every parameter or even what parameters to account for right uh but uh when we want uh when uh as a way to get around this uh problem uh we might want to engage in a proximate basian inference in which uh instead of uh Computing the exact value of a prior we approximate it by doing a kind of margin alization on uh the variables we know the value of it of them and so it kind of gives uh much more um realistic way uh to compute those um those posteriors posteriors uh in any kind of situation so uh that's a much more tractable situation than U or basically we can say that Computing exact basian inference uh in uh most situations can be seen not only intractable but almost impossible um as I said we might not know the exact value of the parameters we want to model so uh yes that's uh one of the um essential issues uh of doing any kind of basing statistics basian inference and um almost always uh the the number here in the parameter in the denominator py is almost always marginalized in um real work situations so that marginalization over py uh instead of the exact value of py um is basically called approximate basian imprints fantastic it was very clear thank you just to add like one more point there because it's a great question so like in the setting of basian phog gentics X the hidden state was the real relationship among species unobserved and then Y is a huge data set so it might be millions of lines or something like that um the likelihood of that data set being observed like Ali said like if you knew that you'd kind of have already known so not only is it a massive data set but but it this question seems to require you to know what you already want to know and then um the variational inference which is a computational heris approach that's existed long before active inference like variational autoencoders and the technique here is kind of like reducing to a certain family of models that only have a few knobs to fiddle with like I'm going to fiddle with the mean and the variance of this distribution and that's it and then there are smoothly incrementally optimizable paths for this sorry for this kind of a problem that you can take in time incrementally versus this which is just like a massive static problem that is intractable for sometimes a number of reasons and so the approach taken to the action perception Loop in active inference just to a first pass is to treat it like this kind of structure fixed though you know asterisk later developments but just at the core level is to treat it like a structurally fixed optimization problem rather than this sort of static number crunch much appreciated thank you um and also as a side note um there's this fascinating pre-print uh which was I think released um a couple of weeks ago namely naturalizing relevance realization so the concept of relevance realization uh by John Veri is uh very relevant here so uh basically what they claim in that paper is uh cognition is fundamentally uh uncomputable in its in uh very well- defined algorithmic sense but uh predictive processing uh via using a probabilistic uh modeling instead of uh using the exact values for parameters uh they can actually provide a much more realistic way uh to in order to model uh the information processing nature of uh any kind of uh biological organisms uh obviously including the brains so I highly recommend reading uh this paper to I think yeah it is this one so yeah so basically relevance realization so what John Veri means by relevance uh is exactly uh the parameters we need to uh model or we need to account for in order to meaningfully model the behavior of the organism uh which is huge okay so any other comments questions or we can look at some of the questions and the coda page all right so uh uh we may dwell a little bit on the distinction between the low road and high road and as career uh opinions uh I mean before reading uh the book for uh those of you uh uh who are approaching this book for the first time uh what do you think uh the advantages or disadvantages of using um each of these approaches uh to active influence literature based on uh this uh preliminary understanding of the distinction between these two approaches or which one would you find more meaningful I don't know more logical and so on I I would say um I was struck by the way that um the high road provides this sort of like um great overview you know like a very elegant explanation that just begins with this simple fundamental predicament of self-organizing systems um so I find it very useful conceptually in that sense I was explaining trying to explain active inference to someone who's not familiar with it recently and uh I was kind of talking about some of these Concepts and he said uh yeah but that's you're you're using mentalizing Concepts right um and uh the a agentic perspective um that's introduced by the high road model I think is interesting and and I'd like to um learn more about how this relates to this sort of um ontology like what the ontology behind it is because it seems to really um go against a lot of the sort of um you know the approach uh the sort of deterministic um uh you know traditional approach where there's this gap between kind of agents and Consciousness and mentalizing Concepts and then a world that's purely determined like is there formal causation because creatures have evolved you know um that's a little mysterious to me so that's that's what struck me great thanks uh yeah uh that's a great point I mean uh one of actually as far as I know one of the main discussion points of whole active inference literature is exactly uh focused on uh kind of unentangled uh what kind of metaphors they use and uh making them more uh precise uh so um I agree that some of some of the metaphors used in active inference can seem at first uh kind of I don't know uh appealing to a kind of panych psychism kind of kind of approach or um I don't know make them uh a bit uh more anth anthropocentric than it than it really is uh but um it's important to keep uh this point in mind that uh some of the terms used in active inference are used in their U very technical and uh mathematically rigorous sense so for instance uh there was a debate around the use of the word sentience uh in one of the papers which published u a few months ago so uh the way active inference literature uses this word sentence is actually uh specifically and rigorously defined in some of the papers uh and it's not necessarily something that perfectly maps onto the everyday conception of of this word or even uh in the sense of sense of the this word that's used in some other relevant disciplines or other related areas uh but yes that's a great point and uh by the way uh I almost forgot it um I usually refer to this great paper or rather essay by Carl priston namely beyond the desert landscape uh from uh the book Andy Clark and his critics in which uh he beautifully uh explains uh the difference between low road and high road and why active inference literature relies more recently on developing the high road approach instead of the low road one so I highly recommend reading uh that essay as well Thomas yeah I mean kind of piggy back on that question on the high road I had a lot of interest in that I don't have experience knowing about Marco blankets I think that seems very clearly one of the major definitions of what we're getting at there so I can't make a comment about that yet I imagine we're going to talk about that in future chapters when I was reading that I reminded from Plato's Mino there there's discourse about the definition of figure and it's using a very particular mode of discussion about to talk about defining what color coming as a followup of figure so and it's a discussion of limit uh in fact even play's character has a very particular kind of nuance discussion that that can even challenge the discussion what limit is so that's is more just meditating on I think that's very I'm interested in hearing more about that um at this point yeah we'll definitely come to uh markup blank get in chapter three and uh hopefully we'll have an extensive discussion around that uh essential concept and I agree markup blanket um I mean holds Central um central place in all active inference literature so uh thanks Daniel Jeff yeah I'm just I'm just reminded uh not from the last bit of conversation but just a little bit before that and I'm sorry I'm jumping in in the middle here uh but I'm just reminded of wheeler's partici for a Universe [Music] um wheeler participant okay so uh interesting uh you see uh I'm not uh very familiar with Wheeler's work but um as far as I know uh I mean some of the ideas from um I mean uh some of Wheeler's ideas and specifically uh the uh bestowing a kind of fundamentality to the concept of information uh which is one of the um wheelers as far as I know um main arguments uh so uh is very very similar to what basian mechanics um way of modeling uh the physics and and I don't mean just physic of belief but actually physics of everything so I'm not sure if you've heard B basian mechanics but it's a kind of new way of uh doing uh even fundamental physics not only uh as a way to model cognitive agents uh but U it is much more fundamental than that uh so uh as the work the works of people like um Delton activ Adel would show uh this new kind of approach to doing uh physics is uh in some sense perfectly congruent with everything uh that has come so far in in the history of fundamental physics in terms of its classical I mean classical physics quantum physics and even for modeling uh dynamical chaotic systems so uh yes um the uh the notion of uh fundamentality of information or rather the fundamentality of beliefs belief in its statistical sense uh which is a kind of uh a mod modulated information is a kind of uh is a kind of information that has a specific probability distribution uh is Central to um I mean uh almost everything active inference related or Fe related so yes I think uh they're closely related to Wheeler's work as well and also obviously Chris Fields a great course on in physics as information processing uh has some great discussions around that point as well I'm editing the journal right now yeah Jeff that will be awesome to see some connections with Chris fields and and to make a comment on this I think cognitive science in a way is looking for that kind of qualified description of things in their environment figure and ground and agent in environment ecological psychology and um there are various like seemingly uh overly simplistic to the point of losing their epistemic or pragmatic value theories like um theories that might be criticized as um like for example even in this discussion words that have been brought up like mentalizing or anthropocentric so clearly those could be taken to a level where it's like yeah it we you know out mentalized where it's not even useful and it's anthropocentric where it doesn't even apply to this or that um or the other way around and then also anything that's part of our behavior and experience it is coming in some way through the apparatus of the person so it can only be escaped so fundamentally and so sometimes there's a movement away from so-called anthropocentrism but then it's like if it doesn't have us doing that theory then what is that really even trying to describe then David yeah hi good morning yes I just wanted to uh see if uh the mous Utopia studies would fall into this dynamic as well uh performed by John Calhoun in the 60s I'm not sure if you're familiar with the studies themselves in terms of uh the behavioral modeling um because they do describe it as a collapse in behavior and in terms of a predetermined you know Utopia from a different experiential point of view and then seeing how that deteriorated despite the constraints on the model itself um wanted to know how that might speak towards uh you know the behavioral modeling itself and uh the collapse function how that might be related to the basium mechanics as well in terms of uh defining an environmental space you know for another U active group of agents for example thanks David I'm not familiar with this work at all so thanks for bringing it up and introducing it um I don't know if anyone is familiar with mous Utopia experiments and uh yeah Jeff go for it no I I that that leads directly into I think um Mar OB Blake and in the holographic principle um and and um layers of separation between agents and the translation that has to occur at every hop right right um for example we're not going to be able to empathize with what it's like to be a bacteria in a bacterial Colony but for the bacteria they take a very bacterio Centric view of the world so I think it's very natural for us and it's right for us to take a anthropocentric uh approach right because that's our lived experience and I think there's definite overlap with with with with psychology and soci the social sciences in general there right um we can't imagine what it's like to be a sun but if you look at tile hard's complexity Theory right like anything with sufficient complexity has Consciousness so you know the sun knows what it's like to be a sun the Sun but but but you know it's building that it's building that translators and the steps between the different systems is is really what what I'm interested in getting after right so it's kind of like the rock being able to imagine it's a rock but not interact with another rock for example yeah yeah yeah exactly yeah I just found it interesting because the uh the mouse St itself where they they teered the different areas and then they kind of segregated it so you know they had you know different levels but you know that's also coming from my background which is sociology so I mean there's a little bit of Prior from myself there as well so so I'd also have to learn more about the experiment specifically but I think one phenomena it speaks to is the unified nature of cognitive phenomena in the developmental in The evolutionary setting like you could study just working memory and you go down the rabbit hole and then working memory is going to bleed off into some other cognitive phenomena because it's all happening in part of this integrated unit of of organismal function and then when we take that view out to like the intergenerational and the social so they're like vertical and horizontal transmission all these different kinds of inheritance mechanisms then oh let's just invent a pencil to offload this and then there's nonlinear effects or let's just make this well make this thing that was clear now let's paint over that like changes to what what can um influence one cognitive phenomena or one kind of intervention or change in the game can lead to like these cascading effects all these different kinds of ecological Dynamics so that kind of motivates that kind of motivates at least the integrated framework to prepare for the um proper analysis of these complex situations yeah do you think those decision non sself motivated in that sense you know where um you know something that's single cellular might you know has two fella for example it's going to drop one um because it may not you know be able to exist with two in that in that you know environment itself but is that necessarily a choice or is that a product of the environment itself and also uh I mean the whole verging field of Basil cognition also deals uh with these kinds of questions I mean whether uh those single celled organisms uh actually possess kind of uh primitive U cognition uh in terms of memory learning decision making and so on and uh how it how it's implemented biologically so uh one uh strong candidate for um for the way I mean for uh the implementation of uh the basil cognition in uh any uh even the single cell organisms uh is through the uh Gap Junction distribution of their of their bioelectric network uh which is the main uh research one of the main areas of research um people such as Michael Levan and his colleagues uh so yeah it's a very interesting um developments in recent uh biology uh so the whole are of Basil cognition as far as I know uh officially started from 2021 so they're very new questions and uh obviously with the mainstream uh biology concerned uh concerned on uh uh the modern synthesis uh approach uh it's a long way uh to somehow nudge uh nudge the Paradigm and uh to make a through true paradigm shift in biology but uh it's happening and uh I I believe uh they can lead to uh some really exciting uh new work in this area once uh we escape uh the traditional uh Gene centered uh view of the organisms uh to the the more holistic and organism centered um ontology so that can that can also apply to uh even non-biological agents if we want to uh think of the Continuum between uh biological things and non-biological ones as a true Continuum not not not uh something that uh that something that has any kind of um fundamental uh difference between between them so basian mechanics uh is one of the approaches that somehow dismantles uh this um discrepan this fundamental discrepancy between uh biological and non-biological agents uh so yeah I believe uh it's a much more interesting worldview to look at things this way in a kind of Continuum as a kind of process and so on instead of uh aristan essentialism that's great I'll just add one more thing I don't want to give too much input but you jogged my memory on something related to something I was thinking about with uh Len's work as well in terms of you know if you have a more macro level like a human being for example and you have uh you know an accident or something and a limb needs to be removed we have this you know Phantom limb we call it where the individual still feels like they can move the limb the limb might still be there but you know it has been removed so obviously there is some memory there but we also know that we have neurons you know and extremities in our bodies you know to what extent is that you know a mechanic of just cognition or is it also a mechanic of the potential energy that was lost in that space for example and would we be able to pinpoint that kind of thing with the technology we have now to try and rep replicate that on a smaller scale to kind of find that you know bioelectric signal or whatever else might be going on in terms of you know the interconnectivity between the micro and the macro and that Dynamic thank you great uh okay Cohen I noticed you had your hand raised if I'm not mistaken you want to add anything or or ywn if I had my hand raised that was a mistake sorry but I I yesterday I listened to the fron and and Levin talk and uh it it seemed super interesting to me that Levin thinks that like he has simulated some systems where like some genotypes understand that other agents share it the genotype and uh they Fair much better and they like communicate much better with agents that are very similar themselves yeah interesting uh actually we'll have a lot more to talk about uh all of these things in the coming chapters and uh as I said it's uh one of the uh main focuses of uh active inference uh especially uh from its high level approach so um I'm excited to hear all the I mean your opinions on all of these matters uh as we come uh to those points as well so Jeff uh did you want to add anything or yeah um I my background is National Security military and uh intelligence community and working overseas like for for the US Military and um there's a massive problem right now with security classification and also with also Enterprise it and cyber security um massive problem with with um information classification controlling classified information and um zero trust and protecting information on the internet right the internet is a scale-free system uh so I think there's a practical applicability here to fundamentally redesigning the way that we look at protecting our information on the internet and this really could be a practicum and I've got about 6,000 words uh in a paper if anybody's interested in collaborating uh short deadline uh I'm targeting it for a conference called the hammeron which is um one of the big National Security conferences in the country it's right outside the NSA headquarters and um yeah so I'm I'm I'm actively working on that um and I I think it's a very very um like re like Salient point and And Timely Point um to our current conversation very interesting thanks for bringing it up if anyone is interested in collaboration uh that would be exciting and great okay so quick question sorry aliot what would be the best way through this group uh to be in touch about things like that in the future I'm not sure Jee if if uh my background would be would be especially helpful on that but I'm just curious about like what's the the sort of best practice on how to link up for that kind of thing if so I'll just answer it um on the cooh six page you can add like a row with your contact info you could also message someone in Zoom if you're synchronously here or just find any other way another option is to write on the projects page and that can include your contact information so if you want to kind of check out other people's projects or plant your own um here if you want to kind of just do it more personally at the this page also understanding that everything that's in the textbook group document is like open essentially and just contact each other and enjoy beyond the halls of The Institute awesome thanks I'm like Curious just any anyone else um even if they haven't added anything like just how is chapter one how did they feel like it opened the book or what did it leave them with yes Petra yeah uh hi I have a question about uh uh normative framework um I'm just thinking is it like uh the framework describes uh healthy functioning organism when we are looking at a human uh as a as a agent or uh is it meant uh differently I'm thinking um in a context of uh Psychopathology when we are talking that the uh agents model generative model is somehow sub optimized uh whether it is connected with this uh normative uh aspect of framework does it make sense sorry great question great question thank you uh so um I'll take a first step so uh uh you see uh the use of this word normative here u in active inference literature and also in the textbook is uh somewhat uh I think defined in the notes if I'm yes it was in page U on page 267 so here it says the term normative means uh there is some evaluative standard against which behavior can be scored active inference is normative in the sense that perception and action are scored by free energy a quantity we will unpack throughout this and the next chapters so as I said again uh it's more technical uh sense of the word normative and it's not necessarily congruent with um other um connotations or meanings of this word so uh basically uh the reason uh it's I mean the word normative used uh in active inference is to refer to uh this Central parameter namely free energy as the main optimization parameter uh around which all the other formulation and um I mean mathematical formalism revolves so uh that's basically uh as far as I understand uh the meaning of the word the adjective normative in active inference literature but I'd be happy to hear any other opinion uh on that matter as well thanks Ali thank you Petra um I'll just add a few other points especially in the social and human there was some great discussion on this question of normativity in the social sciences course with ma alasen um and Ben White and others and AVL last year um so especially in the and um it it it is um an interesting question and one that also people have specifically looked at in the um Psychopathology sense like Carl friston and Thomas par two of the three authors on the textbook are clinicians and so there's a lot of discussion about like well what does it mean if it's Bas optimal but it's DSM diagnosed like how how is it what is it that's going right or wrong what's optimal or nonoptimal and just different ways to talk about like what is optimal like under this um memory parameter this Behavior can be understood as optimal and so that's a kind of like narrow Bas optimal but then that's just what prepares for the discussion like but then the context this is not aligning with their preferences so it's a great question though and and also like um just the is and the art and oh well active inference can be applied to um this system you know we know it can be applied to any system oh can be applied to a company so it ought to be this way but a framework that just describes how things are how do you go from that to how it ought to be and also again as a side note uh I will refer to Mark bier's work as well because uh much of his research uh is um is concentrated on uh again the development of what it actually means uh for U for an organism or an agent to have a normative perception or normative action and uh what it what it means U is a kind of gold Direct edness action or kind of intentional intentional action or as what Terence Deacon would call intentional not necessarily intentional uh so um this kind of normative uh perception or normative action as uh bicker develops in U many of his papers U is the sense that uh this kind of intentionality is uh unseparable from the perception and obviously the action but um in in the so-called um tiop phobic uh community of biologists uh this this kind of discourse uh somehow sometimes would uh raise eyebrows because uh as we know uh from I don't know around U 1950 on word uh there's this kind of phobia around this Theology and any kind of uh I mean go directedness and U to put intentionality uh and mechanistic explanations and so on but uh as Terence ston also shows an incomplete nature um uh and as I said Mark bier's work um this intentionality and normativity uh are essential aspects of uh any sentient and cognitive agent uh and by uh agent I mean anything from single cell organism to uh I don't know much more complex on so um uh thanks for your question yeah oh yeah think great points Allan to kind of connect the like what you described earlier about the Theology and TI tiop phobia and the Deacon and the absenti nature the idea that there are systems um for whom deterministic accounting of their connections accounts for their behavior like a a pendulum or something like that but even a system um like a pendulum could be described as like this absenti movement towards and so there are fundamental parts of especially symbolic knowledge based systems that have um a teely logical cause like the cause of the house is that there was a desire to build a house in that place um and so active inference especially in this chapter one setting it just like brings it all out on the table the chapter does and and the discussions do and um there's a lot in the chapter and also a lot that's kind of like a adjacency that's always fun to bring out and this is kind of just like calibrating to the scope of the kinds of things that that are our starting point for continuing with the journey and just seeing about like some of the heris stics and tools and frames that have been brought um up previously and then in this first half of the textbook we kind of like get through one worked example with the mamalian nervous system by chapter five of like this is kind of of the sketch of what it looks like to start applying this to a system but here's how much work it takes to get to this stage and here's what the model does and doesn't say at that stage thanks Daniel okay so at our final minutes um I'd be interested to hear uh now that you've read chapter one uh what do you anticipate uh or what do you hope uh to gain from Reading um the textbook all through I mean does reading chapter one change your anticipation for better or worse yes as yeah I think Thomas raised his hand oh uh sorry I have some problems seeing the race okay Thomas thanks oh sure um no it hasn't this chapter didn't really change my anticipations mainly because Daniel was so generous when I talked with him in the Discord chat he sent me that paper on order and change in this art basically how to give the active inference account could apply to aesthetic experience phenomenal I mean my research and writings for Plato's account of Aesthetics like lines up perfectly to what that paper was talking about and if anything I'm just more in I'm excited to move forward I think the idea that we I think the account really interesting um I think the fact that priston does a great job making an argument that aesthetic experience is not being suppressed are kind of De decomposed down um basically on the account of epistemic arcs I think that's outstanding um as an account for the aesthetic experience and I think the fact that this can be rigorously discussed is that's the area why I'm even here in the first place because I TR I for anyone doesn't remember call it see meaning me I'm a lecturer in visual art at a university and the biggest challenge I have with pupil students is to rigorously demonstrate how do we know something's more successful or not so this kind of normative measure measure kind of approach um the fact that that can be demonstrated in such rigorous ways I'm very interested um but also the mystery of basically hyra has a whole paper on the question of what's the thing this whole question of That's So the kind of high road even question of how to answer what is a thing uh is just I'm very fascinated nothing's really been changed I'm more just I'm enjoying the meal I'm looking forward to the next course that be my summary so far thanks uh and on the point of Aesthetics uh actually there's this recent um exciting uh special issue um and I think it was in um philosophical transactions of Royal Society on the subject of Aesthetics from the Viewpoint of predictive processing uh which um okay yes Al exactly it was in this issue also tomorrow um with uh Karen Konan the guest stream um about literature and predictive processing that should be super exciting too I'll put this in the notes yeah she uses um the predictive processing uh as applied she applies predictive processing uh to uh literary studies and uh somehow uses literature as a kind of experimental hub for investigating many facets of Consciousness from the Viewpoint of predictive processing it's really fascinating work awesome cool all right well thank you Ali for the facilitation and thank you everyone um next week we'll continue to discuss chapter one in the later time period or feel free to comment this time again and we'll be um talking about chapter 6 the recipe for generative modeling so thank you all farewell thanks for joining everyone bye bye